BioInformatics Research Project: B3DB Model Analysis

Repository: duvvasaivivek/BioInformatics-Research-Project, analysed through the latest commit (f557c47, merge of PR #1, 2026-10-19).

Scope: I cloned the repo and read the full B3DB pipeline, every commit message, and the results and docs. I also ran two checks myself. I did not retrain the models, run the test suite, or read the Initial Model code line by line.

Timeline up to the latest commit

Date	What happened
Aug 13-26	Research paper, extra reference papers and the Zhao 2007 dataset (1,593 of 1,924 compounds) added
Sep 11-18	Initial Model: GINEConv GNN, graph generation, training, prediction, 10 folds, extra test sets
Oct 4-5	Restructure. B3DB pipeline begins: preprocessing, simple model, XGBoost, Optuna, inference script, P-gp model
Oct 6	3D training streamlined, 2D/3D inference decoupled, folders split into BBB/ and Pgp/, old MLP/GNN experiments deleted
Oct 11	Explainability framework, 3D conformer evaluation, manuscript draft
Oct 19	Full XAI suite (SHAP, counterfactuals, applicability domain), validation, ablation and tests, 4,100+ lines in one commit, then merged as PR #1

What the B3DB workflow does

BBB model

1. Data: B3DB classification TSV, 7,807 rows, with 7,805 valid SMILES. 4,956 are BBB+ and 2,849 are BBB-.
2. Features: a 2048-bit Morgan fingerprint (radius 2) plus 6 standardised descriptors: MolWt, LogP, TPSA, HBD, HBA and rotatable bonds.
3. Training: an 80/20 random split with a fixed seed. Optuna runs 50 trials to maximise F1, and the final XGBoost model uses scale_pos_weight, a 2000-tree cap and early stopping.
4. Test result: ROC-AUC 0.954 and accuracy 0.896.

P-gp model

1. Data: Broccatelli via TDC, 1,205 molecules after 3D embedding.
2. Features: the same 2048-bit Morgan fingerprint, plus the same 6 descriptors and 6 RDKit 3D shape descriptors.
3. Training: a stratified split and fixed hyperparameters with no tuning.
4. Test result: ROC-AUC 0.959 and accuracy 0.934.

Layers on top

- inference.py runs both models and puts the molecule into one of four quadrants. For example, BBB+ with P-gp- is the optimal CNS candidate, and BBB+ with P-gp+ is the "paradoxical" Loperamide type.
- TreeSHAP explanations, Morgan-bit-to-substructure mapping, a Tanimoto and leverage applicability domain, a counterfactual analog generator, and HTML/CSV/JSON reports.
- Scripts for validation (1,000 bootstrap resamples), ablation (5-fold CV) and figures. There is also an 8-test regression suite.

The ablation is good scientific practice, and the repo reports its own negative result honestly. 3D shape adds nothing once fingerprints are present: Morgan+2D gets AUC 0.9511, against 0.9474 for the full model with 3D.

Problems found

1. P-gp features differ between training and inference (verified)

- pgp_data_preprocessing.py computes the fingerprint on the hydrogen-added molecule.
- inference.py computes it on the molecule without hydrogens.
- On the stored data, the training fingerprints match the H-added version exactly. Against the inference-style version they differ by about 57 bits per molecule, on a fingerprint with about 45-50 bits on.
- I scored 300 random P-gp molecules with the saved model both ways. Accuracy fell from 0.990 to 0.917 and AUC from 0.996 to 0.983, and 7% of predicted labels flipped. These molecules are mostly training data, so the absolute numbers are inflated, but the shift is real.
- The published test metrics are not affected, because they use the preprocessed CSV. Everything users actually run through inference.py is affected: the quadrant call, the SHAP explanations, the counterfactual claims and the Loperamide demo.
- Fix: compute fingerprints on the molecule without hydrogens in training, or use the H-added molecule at inference. Then retrain and re-run validation.

2. The BBB test set is used for model selection

- Optuna scores trials on the test set, and early stopping also watches it. The 0.954 AUC is therefore somewhat optimistic.
- VALIDATION_PROTOCOL.md claims the split was held out before hyperparameter tuning and before scaler fitting. That is not what the code does.
- Evidence of mild optimism: 5-fold CV with default parameters gives 0.9467, against 0.9544 on the tuned test set.
- A cleaner setup is train / validation / test, or nested CV.

3. Scalers are fit on all the data before splitting

This is minor leakage, but it contradicts the same doc. The BBB split is also not stratified, while the P-gp split is.

4. "Scaffold-aware" evaluation is not a scaffold split

run_validation.py only counts Murcko scaffolds. Both models use random splits, which overstates performance on new chemotypes. A scaffold split would be the honest generalisation test.

5. The manuscript contradicts the CSV

Ablation standard deviations in the manuscript are about half the size of those in ablation_study_results.csv. For example, P-gp Morgan+2D accuracy is ±0.005 in the paper and ±0.0130 in the CSV. The manuscript's BBB std values do not match the CSV either.

6. The README headline overstates 3D

It sells "3D conformation-augmented" as a key capability, but the project's own ablation shows no gain from 3D once Morgan fingerprints are present.

7. Possible leakage in the headline demo (not verified)

Loperamide and the other benchmark drugs may be in the training sets. If they are, the clinical benchmark is not a held-out validation.

8. Smaller items

- bbb_scaler_metadata.json is written by hand. The values do match the data, but the BBB preprocessing script never generates the file.
- The applicability domain samples only about 1,000 of the B3DB molecules.
- The counterfactual example has Tanimoto similarity 0.316, which is low for a "realistic analog".
- os.system('color') at the top of inference.py is Windows-only.
- GetMorganFingerprintAsBitVect is deprecated in newer RDKit.

What to do next

1. Fix the P-gp hydrogen mismatch, then retrain and re-run validation.
2. Redo BBB tuning with a proper validation split, and add a scaffold split for both models.
3. Check whether the benchmark drugs are in the training sets.
4. Regenerate the manuscript tables from the CSVs and correct the validation-protocol doc.

5. Reword the 3D claim as a finding that 3D shape alone gives AUC 0.84 but adds nothing on top of fingerprints.